

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electronics and Computer Engineering) SEMESTER - 3 Winter 2025 (Supply.)

Course :Bachelor of Technology (Electronics and Computer Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: BTECPC302 - ELECTRONICS DEVICES & CIRCUITS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 The JFET is a _____ controlled device.
 - A. Current
 - B. Voltage
 - C. Power
 - D. None
 - 2 Pinch-off occurs when:
 - A. Drain current becomes zero
 - B. Channel is completely cut off
 - C. Drain current becomes constant
 - D. Gate draws current
 - 3 In an n-channel E-MOSFET, the channel is formed by:
 - A. Doping
 - B. Inversion
 - C. Diffusion
 - D. Pinch off
 - 4 The input stage of an op-amp is usually a:
 - A. Common emitter stage
 - B. Differential amplifier
 - C. Push-pull stage
 - D. Class C power stage
 - 5 A voltage follower has a gain of:
 - A. Greater than 1
 - B. Less than 1
 - C. Exactly 1
 - D. Depends on resistors

- 6 An ideal integrator uses:
 - A. Capacitor at input and resistor in feedback
 - B. Resistor at input and capacitor in feedback
 - C. Inductor in input
 - D. Transformer in feedback
- 7 Voltage-series feedback results in:
 - A. High input impedance
 - B. Low input impedance
 - C. High output impedance
 - D. Decreased bandwidth
- 8 With negative feedback, bandwidth becomes:
 - A. Smaller
 - B. Larger
 - C. Same
 - D. Zero
- 9 Total gain of cascade amplifier is:
 - A. Sum of stage gains
 - B. Difference
 - C. Average of stage gains
 - D. Product of stage gains
- 10 A discrete transistor voltage regulator mainly uses:
 - A. Transistor + Zener diode
 - B. MOSFET + capacitor
 - C. Inductor + resistor
 - D. Relay + diode
- 11 Major disadvantage of a shunt regulator is:
 - A. Complex design
 - B. High efficiency
 - C. Low efficiency (current always flows)
 - D. No voltage regulation
- 12 Thermistors are based on:
 - A. Piezoelectric effect
 - B. Change of resistance with temperature
 - C. Magnetic induction
 - D. Optical absorption

Q2. Solve the following.

- A) Draw and explain the circuit arrangement for VI characteristics of N-channel JFET and also draw the characteristic by showing the different regions. 6
- B) A JFET with parameters $I_{DSS} = 12 \text{ mA}$, $V_{(GS(off))} = -6 \text{ V}$, that provides $V_{GS} = -3 \text{ V}$. Determine the value of drain current I_D . 6

Q3. Solve the following.

- A) Explain the principle of MOSFET in enhancement mode with neat sketches and also draw the output (drain) characteristics. 6
- B) Derive the differential mode gain of a dual-input balanced-output differential amplifier. 6

Q4. Solve Any Two of the following.

- A) Explain the inverting and non-inverting amplifier configurations with gain expressions. 6
 - B) Explain the working of an Op-amp integrator with proper equations. 6
 - C) Explain the concept of feedback in amplifiers and describe the four feedback topologies with diagrams. 6
- Q5. Solve Any Two of the following.
- A) Describe the working of an RC phase-shift oscillator and derive formula for frequency of oscillation. 6
 - B) Describe the construction and operation of a three-terminal 78xx series IC voltage regulator. 6
 - C) Explain the working of a shunt voltage regulator and list its advantages and disadvantages. 6
- Q6. Solve Any Two of the following.
- A) A discrete transistor voltage regulator uses a Zener diode of 8.2 V as reference. The transistor has $V_{BE} = 0.7$ V. If the load draws 100 mA, find the regulated output voltage and the power dissipated in the transistor when $V_{in} = 15$ V. 6
 - B) State and explain the essential requirements of a good sensor/transducer. 6
 - C) Explain the working principle of an LVDT (Linear Variable Differential Transformer) with neat diagram. 6

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